Chapter

3

ALGEBRA

U8T5D2

Learning Objectives

- To understand the following identities through geometrical proof
  - $(x+a)(x+b)=x^2+x(a+b)+ab$
  - $(a+b)^2=a^2+2ab+b^2$
  - $(a-b)^2=a^2-2ab+b^2$ and
  - $(a+b)(a-b)=a^2-b^2$.
- Able to apply the identities in numerical problems.
- Able to recognise expressions that are factorisable.
- To represent inequalities in one variable, graphically.

3.1 Introduction to Identities

In earlier classes, we have learnt to construct algebraic expressions using exponential notations. For example, x^2+3x+2 is an algebraic expression in the variable x. This can also be written as an equation $x^2+3x=-2$.

By substituting numerical values for x, we can verify this equation.

If $x=-2$, then $\quad$ L.H.S $=x^2+3x=(-2)^2+3(-2)$

$=4-6$

$=-2=$ R.H.S

Hence, this equation is true when $x=-2$.

If $x=-1$, then $\quad$ L.H.S $=x^2+3x=(-1)^2+3(-1)$

$=1-3$

$=-2=$ R.H.S

Hence, this equation is true when $x=-1$.

But when $x=1$, $\quad$ L.H.S $=x^2+3x=(1)^2+3(1)$

$=1+3$

$=4\neq$ R.H.S

Thus, this equation is not true when $x=1$.

Thus, $x^2+3x=-2$ is an equation which is true only when x takes the values -1 and -2. Hence, an equation is true only for certain values of the variable in it.

It is not true for all values of the variables.

Now, consider the algebraic expression, $(a+b)^2=a^2+2ab+b^2$. Let us try to find the values of the expression for the given values of a and b.